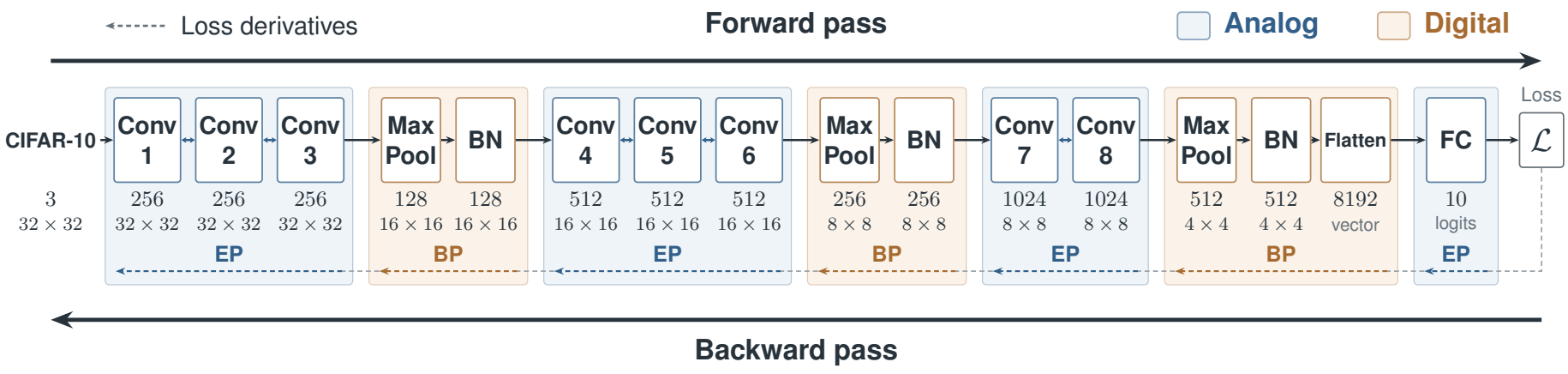

Loss derivatives
Forward pass
Analog
Digital
CIFAR-10
3
32 × 32
Conv 1
256
32 × 32
Conv 2
256
32 × 32
Conv 3
256
32 × 32
EP
Max Pool
128
16 × 16
BN
128
16 × 16
BP
Conv 4
512
16 × 16
Conv 5
512
16 × 16
Conv 6
512
16 × 16
EP
Max Pool
256
8 × 8
BN
256
8 × 8
BP
Conv 7
1024
8 × 8
Conv 8
1024
8 × 8
EP
Max Pool
512
4 × 4
BN
512
4 × 4
Flatten
8192
vector
BP
FC
10
logits
EP
Loss
$\mathcal{L}$
Backward pass